Programming Language Fundamentals
Oregon State University Cascades
Spring 2026

Bug Hunt 1: The Stack

What the program does

`stack.erl` defines a **stack ADT** implemented as a chain of 2-tuples. The empty stack is the atom `empty`. A non-empty stack with top value `V` and rest `R` is the tuple {`V, R`}. So the stack containing `[a, b, c]` (with `a` on top) is {`a,` {`b,` {`c, empty`}}}.

The module exports:

- `new(List)` — build a stack from a list (head of list is top of stack).
- `push(Stack, Value)` — return a new stack with `Value` on top.
- `pop(Stack)` — return {`Value, NewStack`} for the top element.
- `get(Stack, I)` — return the value at 0-indexed position `I`.
- `set(Stack, I, NewValue)` — return a new stack with position `I` replaced.
- `remove(Stack, I)` — return {`RemovedValue, NewStack`}.
- `print_td(Stack)` — print elements **top-down** (top of stack first).
- `print_bu(Stack)` — print elements **bottom-up** (bottom of stack first).

`set`, `get`, and `remove` all use the **helper-stack pattern**: walk toward index `I` while pushing each element onto a helper stack, then rebuild the result by popping the helper back onto a new stack.

`main.erl` is a driver that exercises every operation.

The catch

This program contains **exactly 3 bugs** that prevent it from producing the expected output. The bugs are all *semantic* — the program compiles without errors and runs to completion without crashing.

You may not run the program until you have submitted your answer. Trace by reading.

Expected output

```
popped: cooper
pushed: matthew
get(2): ashley
set(2, kameron)
popped: matthew
popped: elizabeth
```

```
Top-down:
kameron
austin
dalton
tristan
daniel

Bottom-up:
daniel
tristan
dalton
austin
kameron

Fresh stack top-down:
one
two
three
Fresh stack bottom-up:
three
two
one
```

Your write-up

For each bug, record:

1. **Location** — file name and line number(s).
2. **Diagnosis** — in 1–2 sentences, what is wrong.
3. **Fix** — the corrected line(s).
4. **Symptom** — which line(s) of the expected output differ from what the buggy program actually prints, and why this specific bug causes that specific symptom.